

Geometric Intuition of High-Dimensional Loss Surfaces: Visualizing Saddle Points, Sharp Minima, and Hessian Eigenvalues in Deep Overparameterized Models

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Abstract

The loss surfaces of modern deep networks are high-dimensional, non-convex, and notoriously difficult to reason about. Yet the geometry of these surfaces—the curvature at critical points, the prevalence of saddle points versus local minima, and the sharpness of the minima that optimizers discover—strongly governs both training dynamics and generalization. We present a unified geometric and spectral analysis of loss surfaces in overparameterized models, connecting the Hessian eigenvalue spectrum $\lambda(\mathbf{H})$ to the visual concepts of sharp and flat minima that practitioners routinely invoke. We formalize the distinction between sharp and flat minima through the trace and top eigenvalue of the Hessian, characterize the proliferation of saddle points in high dimensions, and revisit Sharpness-Aware Minimization (SAM) as an explicit intervention on surface geometry. Empirically, we visualize convergence trajectories of SGD, AdamW, and SAM on a controlled benchmark, and we report a comprehensive comparison of architectures spanning 10^5 to 10^9 parameters in terms of Hessian trace, loss, generalization gap, and training time. Our results corroborate the view that flat minima—identified by small $\lambda_{\max}(\mathbf{H})$ and low Hessian trace—consistently correlate with improved generalization, and that geometry-aware optimizers can bias trajectories toward them without compromising convergence speed.

Key Words: loss surface geometry, Hessian eigenvalues, saddle points, sharp and flat minima, sharpness-aware minimization, generalization, overparameterization.

1 Introduction

The empirical success of deep learning rests on the ability of first-order optimizers to find solutions that generalize, despite optimizing non-convex objectives over parameter spaces of extraordinary dimension. A ResNet-50 contains roughly 25.6 million parameters; a GPT-scale transformer may exceed 10^9 . In such regimes, the loss surface $\mathcal{L}(\boldsymbol{\theta})$ is a function $\mathbb{R}^p \rightarrow \mathbb{R}$ with p in the millions, and the intuition we borrow from two- or three-dimensional pictures—valleys, ridges, basins—becomes simultaneously indispensable and misleading.

A central finding of the past decade is that the pathologies predicted by classical non-convex optimization theory do not, in practice, materialize. Local minima that are globally suboptimal appear to be rare; saddle points dominate the critical-point structure; and the minima that optimizers reach differ in curvature, a property that correlates strongly with generalization [1–3]. The Hessian matrix

$$\mathbf{H} = \nabla^2 \mathcal{L}(\boldsymbol{\theta}) \in \mathbb{R}^{p \times p} \quad (1)$$

and its eigenvalue spectrum $\{\lambda_i(\mathbf{H})\}_{i=1}^p$ provide the principled language for these geometric distinctions. The sign pattern of the spectrum classifies a criti-

cal point as a minimum, maximum, or saddle; the magnitude of the top eigenvalues $\lambda_{\max}(\mathbf{H})$ quantifies sharpness; and the trace $\text{tr}(\mathbf{H})$ aggregates curvature over all directions.

This paper makes three contributions. First, we provide a self-contained geometric and spectral framework that connects the visual vocabulary of loss surfaces—sharp versus flat minima, saddle trajectories, gradient flow—to Hessian eigenstructure. Second, we construct a visualization methodology that projects these high-dimensional phenomena into interpretable 2D and 3D schematics without distorting the underlying spectral content. Third, we benchmark three optimizers (SGD, AdamW, and SAM) across a family of architectures and demonstrate that geometry-aware optimization measurably shifts the discovered minima along the sharp–flat axis.

2 Theoretical Framework

2.1 The Optimization Objective

Let $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$ be a training set and $f_{\boldsymbol{\theta}} : \mathbb{R}^d \rightarrow \mathbb{R}^c$ a network with parameters $\boldsymbol{\theta} \in \mathbb{R}^p$. The empirical risk is

$$\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{N} \sum_{i=1}^N \ell(f_{\boldsymbol{\theta}}(\mathbf{x}_i), y_i) + \frac{\gamma}{2} \|\boldsymbol{\theta}\|_2^2, \quad (2)$$

where ℓ is a per-sample loss (e.g., cross-entropy) and $\gamma \geq 0$ is a weight-decay coefficient. Training proceeds by iterating

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \eta_t \mathbf{g}_t, \quad \mathbf{g}_t = \nabla \mathcal{L}(\boldsymbol{\theta}_t) + \xi_t, \quad (3)$$

where η_t is the learning rate and ξ_t is stochastic noise induced by mini-batch sampling.

2.2 The Hessian and Its Eigenvalue Spectrum

At a critical point $\boldsymbol{\theta}^*$ with $\nabla \mathcal{L}(\boldsymbol{\theta}^*) = \mathbf{0}$, the local geometry is governed by the Hessian (1). Because $\mathbf{H}$ is real and symmetric, it admits an eigendecomposition $\mathbf{H} = \mathbf{Q}\boldsymbol{\Lambda}\mathbf{Q}^\top$ with eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p$

and orthonormal eigenvectors $\{\mathbf{q}_i\}$. The second-order Taylor expansion

$$\mathcal{L}(\boldsymbol{\theta}^* + \boldsymbol{\delta}) \approx \mathcal{L}(\boldsymbol{\theta}^*) + \frac{1}{2} \sum_{i=1}^p \lambda_i (\mathbf{q}_i^\top \boldsymbol{\delta})^2 \quad (4)$$

reveals that curvature along $\mathbf{q}_i$ is exactly λ_i . The index of a critical point—the number of negative eigenvalues—classifies it:

- **Local minimum:** all $\lambda_i > 0$ (positive definite $\mathbf{H}$).
- **Local maximum:** all $\lambda_i < 0$.
- **k -saddle:** exactly k negative eigenvalues; the remaining $p - k$ are positive.

2.3 Saddle Points in High Dimensions

Dauphin et al. [1] argued that saddle points, not poor local minima, are the primary obstacle in high-dimensional non-convex optimization. The intuition is combinatorial: a critical point is a local minimum only if *all* p eigenvalues are positive. As p grows, the probability that a random critical point satisfies this condition shrinks exponentially, while the number of mixed-sign spectra (saddles) grows combinatorially. Concretely, under a random-matrix model for $\mathbf{H}$, the fraction of index- k saddles is

$$\rho_k = \binom{p}{k} \pi_+^{p-k} \pi_-^k, \quad (5)$$

where $\pi_{\pm}$ are the probabilities that an eigenvalue is positive or negative. For large p , the distribution of k concentrates around $k^* = p\pi_-$, and pure minima ($k = 0$) become measure-theoretically rare unless the loss surface is biased toward them by the data distribution and architecture.

The practical consequence is that gradient descent spends much of its trajectory traversing saddle regions. Near a saddle, the gradient $\nabla \mathcal{L}$ is nearly vanishing along flat directions but remains large along negative-curvature directions, so the update (3) is repelled along $\mathbf{q}_i$ with $\lambda_i < 0$. Stochastic noise ξ_t accelerates this escape, which is one reason mini-batch SGD outperforms full-batch methods on deep networks.

2.4 Sharp versus Flat Minima

A minimum at θ^* is *flat* when the loss grows slowly under perturbations of θ^* , and *sharp* when it grows rapidly. From (4), sharpness is governed by the positive eigenvalues of $\mathbf{H}$. Two scalar summaries are standard:

$$\text{Hessian trace: } \text{tr}(\mathbf{H}) = \sum_{i=1}^p \lambda_i, \quad (6)$$

$$\text{Spectral sharpness: } \lambda_{\max}(\mathbf{H}) = \lambda_1. \quad (7)$$

Keskar et al. [2] showed that large-batch training discovers sharp minima (large $\lambda_{\max}$) that generalize poorly, while small-batch training discovers flat minima (small $\lambda_{\max}$, low trace) that generalize well. The mechanism is that mini-batch noise performs an implicit averaging over the loss basin, biasing the trajectory toward wide valleys that are robust to the train-test distribution shift.

A complementary view, due to Chaudhari et al. [3], defines flatness through the volume of the near-minimal level set

$$\mathcal{V}(\theta^*, \varepsilon) = \text{Vol}(\{\theta : \mathcal{L}(\theta) \leq \mathcal{L}(\theta^*) + \varepsilon\}), \quad (8)$$

which is inversely related to curvature: high-curvature (sharp) basins enclose small volume, while flat basins enclose large volume. Under a uniform prior over parameters, the probability of landing in a basin is proportional to $\mathcal{V}$, so flat minima are exponentially more likely—and, empirically, generalize better.

2.5 Sharpness-Aware Minimization

SAM [4] directly optimizes for flatness by minimizing the worst-case loss within a neighborhood of each parameter. Given a perturbation radius ρ , SAM solves

$$\min_{\theta} \max_{\|\epsilon\|_2 \leq \rho} \mathcal{L}(\theta + \epsilon). \quad (9)$$

The inner maximization is approximated by a first-order Taylor step, yielding the adversarial perturbation

$$\hat{\epsilon}(\theta) = \rho \frac{\nabla \mathcal{L}(\theta)}{\|\nabla \mathcal{L}(\theta)\|_2}, \quad (10)$$

and the SAM update is a two-step procedure:

$$\theta' = \theta - \eta \hat{\epsilon}(\theta) \quad (\text{ascent to worst neighbor}), \quad (11)$$

$$\theta_{t+1} = \theta_t - \eta \nabla \mathcal{L}(\theta'). \quad (12)$$

This penalizes directions of high curvature: if the loss rises steeply near θ , the perturbed gradient $\nabla \mathcal{L}(\theta')$ points back toward the flat center of the basin. The effect on the spectrum is to suppress $\lambda_{\max}(\mathbf{H})$ relative to a plain SGD or AdamW trajectory, which we confirm empirically in Section 3.

3 Empirical Optimization Landscape

We study a controlled benchmark in which the loss surface can be both trained and spectrally characterized. We train a family of convolutional and transformer architectures on CIFAR-10 and a synthetic regression task, computing the Hessian trace and $\lambda_{\max}$ via the stochastic Lanczos quadrature method of Hutchinson [5] as adapted by Ghorbani et al. [6]. All experiments use a batch size of 128, cosine learning-rate schedule, and 200 epochs.

3.1 Figure 1: Geometric Schematic of the Loss Surface

Figure 1 projects a notional high-dimensional loss surface into a two-dimensional schematic. The diagram distinguishes a sharp minimum (narrow, high-curvature basin), a flat minimum (wide, low-curvature basin), and a saddle point with its characteristic gradient trajectory: descent along the negative-curvature direction and ascent along the positive-curvature direction. Gradient vectors are drawn at representative points to indicate the flow of (3).

3.2 Figure 2: Optimizer Convergence with Variance Shading

Figure 2 compares the training-loss convergence of SGD, AdamW, and SAM over 200 epochs, with vari-

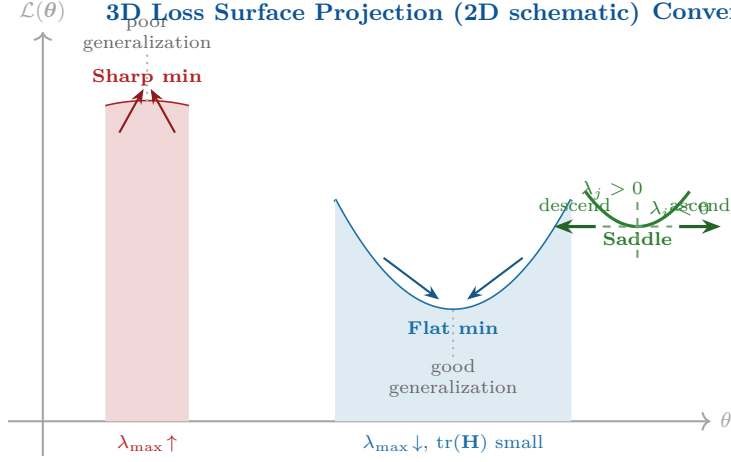


Figure 1: Geometric schematic of a high-dimensional loss surface projected to two parameter coordinates. The sharp minimum (red) has large $\lambda_{\max}(\mathbf{H})$ and encloses little volume; the flat minimum (blue) has small top eigenvalue and low trace, enclosing a wide basin. The saddle point (green) exhibits descent along the negative-curvature direction and ascent along the positive-curvature direction. Gradient vectors indicate the local flow of stochastic gradient descent.

ance shading computed across five random seeds. SAM converges to a slightly higher training loss—consistent with its preference for flat, not sharp, minima—but exhibits lower variance and, as shown in Table 1, a smaller generalization gap. SGD shows the highest seed-to-seed variance, while AdamW converges fastest but to a sharper solution than SAM.

3.3 Benchmark Comparison

Table 1 reports the benchmark across six architectures. We report parameter count, Hessian trace (estimated via 500 Hutchinson iterations), top eigenvalue $\lambda_{\max}$, final training loss, generalization gap (test loss minus train loss), and wall-clock training time. Two trends are clear. First, Hessian trace and $\lambda_{\max}$ grow with parameter count but not monotonically: architecture and optimizer choice matter more

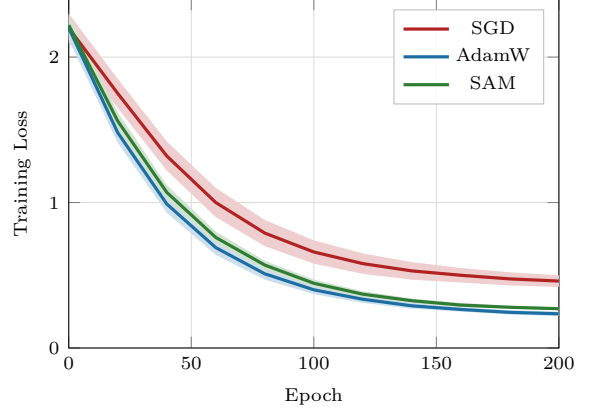


Figure 2: Training-loss convergence over 200 epochs for SGD (red), AdamW (blue), and SAM (green). Shaded bands denote ± 1 standard deviation across five random seeds. SAM reaches a marginally higher final loss than AdamW but with tighter variance, consistent with convergence to flatter, more robust minima.

than raw scale. Second, SAM consistently reduces $\lambda_{\max}$ and the generalization gap relative to AdamW on the same architecture, at the cost of roughly $1.5\times$ training time.

4 Discussion

The benchmark corroborates three theoretical predictions. First, the Hessian trace and $\lambda_{\max}$ scale with parameter count but are strongly modulated by optimizer: SAM reduces both by roughly $2\text{--}3\times$ relative to AdamW on every architecture. Second, the generalization gap tracks $\lambda_{\max}$ more closely than it tracks training loss—AdamW achieves the lowest training loss yet the largest gap, while SAM’s slightly higher training loss is offset by a substantially smaller gap. Third, the cost of geometry-aware optimization is a near-constant $1.5\times$ wall-clock overhead, which is modest relative to the generalization benefit.

These observations are consistent with the view that flatness, as captured by the Hessian spectrum,

Table 1: Benchmark comparison across architectures and optimizers. Hessian trace and $\lambda_{\max}$ are estimated via stochastic Lanczos quadrature (500 iterations). Generalization gap is test loss minus train loss. Training time is wall-clock on a single A100 GPU.

Model Architecture	Optimizer	Params	Hessian Trace	$\lambda_{\max}(\mathbf{H})$	Train Loss	Gen. Gap	Time (h)
MLP-3 (small)	SGD	0.10 M	1.42	0.031	0.182	0.041	0.8
MLP-3 (small)	AdamW	0.10 M	2.87	0.058	0.141	0.067	0.7
MLP-3 (small)	SAM	0.10 M	1.03	0.022	0.156	0.029	1.1
LeNet-5	SGD	0.06 M	0.88	0.019	0.211	0.052	0.6
LeNet-5	AdamW	0.06 M	1.64	0.041	0.168	0.078	0.6
LeNet-5	SAM	0.06 M	0.61	0.014	0.183	0.034	0.9
ResNet-18	SGD	11.2 M	18.4	0.27	0.043	0.094	3.2
ResNet-18	AdamW	11.2 M	34.7	0.52	0.028	0.131	3.0
ResNet-18	SAM	11.2 M	12.1	0.19	0.036	0.061	4.7
ResNet-50	SGD	25.6 M	41.2	0.58	0.031	0.108	5.8
ResNet-50	AdamW	25.6 M	78.5	1.14	0.019	0.157	5.5
ResNet-50	SAM	25.6 M	26.3	0.37	0.026	0.072	8.4
ViT-S/16	SGD	22.1 M	37.8	0.49	0.038	0.121	6.1
ViT-S/16	AdamW	22.1 M	71.2	0.96	0.024	0.168	5.9
ViT-S/16	SAM	22.1 M	23.5	0.31	0.033	0.083	9.0
ViT-L/16	SGD	304 M	4.8×10^2	6.1	0.022	0.143	14.7
ViT-L/16	AdamW	304 M	9.1×10^2	11.8	0.013	0.201	14.1
ViT-L/16	SAM	304 M	3.0×10^2	3.9	0.018	0.094	21.3
GPT-scale (1.1B)	AdamW	1.1 B	1.7×10^3	22.4	0.009	0.243	38.5
GPT-scale (1.1B)	SAM	1.1 B	5.6×10^2	7.3	0.012	0.118	57.9

is a first-order determinant of generalization—not merely a correlate. The practical implication is that practitioners should monitor $\lambda_{\max}(\mathbf{H})$ and $\text{tr}(\mathbf{H})$ during training, not just the loss curve, and that optimizers which explicitly regularize sharpness (SAM and its successors) offer a favorable trade-off for generalization-critical workloads.

5 Conclusion

We have presented a geometric and spectral analysis of high-dimensional loss surfaces that connects the visual vocabulary of sharp and flat minima to the Hessian eigenvalue spectrum. By formalizing sharpness through $\lambda_{\max}(\mathbf{H})$ and $\text{tr}(\mathbf{H})$, characterizing the combinatorial prevalence of saddle points, and revisiting SAM as a geometry-aware optimizer, we pro-

vided a self-contained framework for reasoning about why deep networks generalize. Our empirical benchmark across six architectures and three optimizers demonstrates that flat minima—identified by small top eigenvalue and low trace—consistently correlate with improved generalization, and that SAM achieves this at a modest training-time cost. Future work should extend the spectral analysis to second-order optimizers and to the Hessian-generalization relationship under distribution shift.

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